

Q1: 1. $x \in \mathbb{C}^n$, $y \in \mathbb{C}^n$ ($a+ib$
 $i = \sqrt{-1}$)

$$x = \begin{bmatrix} 3-2i \\ -2+i \\ -4-3i \end{bmatrix} \in \mathbb{C}^3, \quad y = \begin{bmatrix} -2+4i \\ 5-i \\ -2i \end{bmatrix} \in \mathbb{C}^3$$

let $z = x + y$. Compute z and $\bar{z}$.

Soln - $z = x + y = \begin{bmatrix} 1+2i \\ 3 \\ -4-5i \end{bmatrix}$

$$\bar{z} = \begin{bmatrix} 1-2i \\ 3 \\ -4+5i \end{bmatrix}$$

Q(2) Compute $x \cdot y = \bar{x}^T y$ ($x^T y \rightarrow$ in real numbers)

Soln:-

2.

$$x = \begin{bmatrix} 3-2i \\ -2+i \\ -4-3i \end{bmatrix}, \quad y = \begin{bmatrix} -2+4i \\ 5-i \\ -2i \end{bmatrix}$$

$$\bar{x} = \begin{bmatrix} 3+2i \\ -2-i \\ -4+3i \end{bmatrix}; \quad \bar{x}^T = \begin{bmatrix} 3+2i & -2-i & -4+3i \end{bmatrix}$$

$$\bar{x}^T y = \begin{bmatrix} 3+2i & -2-i & -4+3i \end{bmatrix}_{1 \times 3} \begin{bmatrix} -2+4i \\ 5-i \\ -2i \end{bmatrix}_{3 \times 1}$$

$$= (3+2i)(-2+4i) + (-2-i)(5-i) + (-4+3i)(-2i)$$

$$= (8i-14) + (-11-3i) + (6+8i)$$

$$= (-14-11+6) + (8-3+8)i$$

$$= \boxed{-19 + 13i} \quad (\text{Ans}).$$

Q.3'. let $A = \begin{pmatrix} 1/\sqrt{2} & (1+i)/\sqrt{2} \\ i/\sqrt{2} & (1-i)/\sqrt{2} \end{pmatrix}$

3.

Find the length of the column vectors.

Soln:-

$$C_1 = \begin{bmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{bmatrix}, \quad C_2 = \begin{bmatrix} (1+i)/\sqrt{2} \\ (1-i)/\sqrt{2} \end{bmatrix}$$

length of C_1 :

$$\begin{aligned} \|C_1\| &= \sqrt{\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{i}{\sqrt{2}} \right|^2} \\ &= \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1. \end{aligned}$$

length of C_2 :

$$\begin{aligned} \|C_2\|^2 &= \overline{C_2}^T C_2 \\ &= \begin{pmatrix} \frac{1-i}{\sqrt{2}} & \frac{1+i}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1+i}{\sqrt{2}} \\ \frac{1-i}{\sqrt{2}} \end{pmatrix} \end{aligned}$$

4.

$$= \left(\frac{1-i}{\sqrt{2}} \right) \cdot \left(\frac{1+i}{\sqrt{2}} \right) + \left(\frac{1+i}{\sqrt{2}} \right) \cdot \left(\frac{1-i}{\sqrt{2}} \right)$$

$$= \left(\frac{1-i^2}{2} \right) + \left(\frac{1-i^2}{2} \right)$$

$$= 2 \cdot \left(\frac{1-i^2}{2} \right)$$

$$= 2.$$

$$A^* = \bar{A}^T = A^T$$

↙
A is real
symm.

$$\bar{A}^T = A \rightarrow \text{Hermitian matrices.}$$

$$A^* = A.$$

proof:-

$$Av = \lambda v$$

$$\Rightarrow \lambda v = Av$$

$$v^* (\lambda v) = v^* Av$$

$$\begin{aligned} \lambda (v^* v) &= v^* Av \\ &= v^* \downarrow A^* v \\ &= (Av)^* v \end{aligned}$$

$$\underbrace{A^* = A}$$

$$(AB)^T = B^T A^T$$

$$= (dv)^* v$$

5.

$$d(v^* v) = \bar{d} v^* v$$

$$\Rightarrow (v^* v)(d - \bar{d}) = 0$$

$$\underbrace{v^* v = 0}_{\text{Yes or No}} \quad \text{Yes or No } (\text{X})$$

$$\boxed{d = \bar{d}}$$

Q4: Which among the following matrices are Hermitian?

$$A = \begin{bmatrix} 2i & 3+3i \\ 3-3i & 5i \end{bmatrix} \rightarrow \text{not Hermitian}$$

$$B = \begin{bmatrix} 2 & 3+3i \\ 2-3i & 5 \end{bmatrix} \rightarrow \text{not Hermitian}$$

$$C = \begin{bmatrix} \textcircled{2} & 3-3i \\ 3+3i & \textcircled{5} \end{bmatrix} \rightarrow \text{Hermitian}$$

6.

Q.50:- $A = \begin{bmatrix} 2 & 3-3i \\ 3+3i & 5 \end{bmatrix}$. Find the

complex eigenvector for the eigenvalue $\lambda = 8$.

Soln:- $(A - \lambda I)v = 0$

$\Rightarrow (A - 8I)v = 0$

$\Rightarrow \begin{bmatrix} 2-8 & 3-3i \\ 3+3i & 5-8 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} -6 & 3-3i \\ 3+3i & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-6v_1 + (3-3i)v_2 = 0 \quad \text{--- (1)}$

$(3+3i)v_1 - 3v_2 = 0 \quad \text{--- (2)}$

$(1+i)(3+3i)v_1 - (-1+i)3v_2 = 0$

$\Rightarrow -6v_1 + (3-3i)v_2 = 0 \quad \text{--- (3)}$

$$\Rightarrow -\cancel{6} v_1 = \frac{(3-3i)}{6} v_2$$

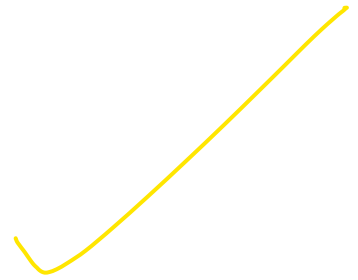
7.

$$v_1 = \left(\frac{1-i}{2} \right) \cdot v_2$$

$$\underline{v_2 = 2},$$

$$v_1 = (1-i)$$

$$v = \begin{bmatrix} 1-i \\ 2 \end{bmatrix}$$



Q.6:

$$A = \begin{bmatrix} \textcircled{1} & 1-i & 3-2i \\ 1+i & \textcircled{-5} & -i \\ 3+2i & i & \textcircled{0} \end{bmatrix}$$

→ whether eigenvalues are real or ~~complex~~?

soln

$$\underline{\overline{A}^T = A}$$

Unitary Matrices : ① U has orthonormal columns.

8.

② $U^H U = I$

③ $U U^H = I$

④ $U^{-1} = U^H$

$$Q \rightarrow \underline{Q^T Q = I}$$

$$\underline{Q Q^T = I}$$

$$\underline{Q^{-1} = Q^T}$$

Q. let $A = \frac{1}{2} \begin{bmatrix} 1+i & \sqrt{k} \\ 1-i & \sqrt{k}i \end{bmatrix}$, where $k \in \mathbb{R}$.

Find the value of k such that A is unitary.

Soln. - For a unitary matrix A , its columns must be of unit length.

$$C_2 = \frac{1}{2} \begin{bmatrix} \sqrt{k} \\ \sqrt{k}i \end{bmatrix}. \quad \text{Then, } \|C_2\| = 1$$

$$\Rightarrow \|c_2\|^2 = 1^2 = 1$$

$$c_2 = \frac{1}{2} \begin{bmatrix} \sqrt{k} \\ \sqrt{k}i \end{bmatrix}$$

$$\Rightarrow \|c_2\|^2 = c_2^* c_2 = \bar{c}_2^T c_2$$

$$= \frac{1}{2} \begin{bmatrix} \sqrt{k} & (-\sqrt{k}i) \end{bmatrix} \cdot \frac{1}{2} \begin{bmatrix} \sqrt{k} \\ (\sqrt{k}i) \end{bmatrix}$$

$$= \frac{1}{4} [k + k]$$

$$= \frac{2k}{4}$$

$$\|c_2\|^2 = \frac{k}{2}$$

$$\Rightarrow \|c_2\|^2 = 1 \Rightarrow \frac{k}{2} = 1$$

$$\Rightarrow \boxed{k = 2}$$

Q:

10.

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

is unitary or not??

$$\vec{c}_1^T \cdot \vec{c}_2 = 0$$

c_1

c_2

Not unitary matrix.

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \frac{i}{2} + \frac{i}{2} = \frac{2i}{2} = i \neq 0.$$

Every matrix cannot be diagonalized. What to do then?

Any $n \times n$ matrix A can be diagonalized as!

$$A = U \left(\sum_{i=1}^n \right) V^T$$

orthogonal diagonal

orthogonal

SVD: Singular value decomposition.

Q.1.

Find the SVD for the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}.$$

11.

Solution:-

Step 1:- compute $A^T A$.

$$A^T A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix}$$

Step 2:- Find the eigenvalues of $A^T A$.

$$|A^T A - \lambda I| = \begin{vmatrix} 10 - \lambda & 20 \\ 20 & 40 - \lambda \end{vmatrix}$$

$$= (10 - \lambda)(40 - \lambda) - 400$$

$$= 400 - 10\lambda - 40\lambda + \lambda^2 - 400$$

$$= \lambda^2 - 50\lambda$$

$$\Rightarrow \lambda^2 - 50\lambda = 0$$

$$\Rightarrow \lambda(\lambda - 50) = 0$$

11

$$; \boxed{\lambda_1 = 50, \lambda_2 = 0}$$

Step 3:-

Find the singular values.

12.

$$\sigma_i = \sqrt{d_i}$$

So,

$$\sigma_1 = \sqrt{50} = 5\sqrt{2}, \quad \sigma_2 = 0$$

$$\Sigma = \begin{bmatrix} 5\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix}$$

Step 4:- Find the right singular vectors (columns of V).

$$\rightarrow (A^T A - d I) v = 0.$$

$$A^T A = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix}$$

For $d_1 = 50$

$$\begin{bmatrix} 10-50 & 20 \\ 20 & 40-50 \end{bmatrix} v = 0$$

$$\Rightarrow \begin{bmatrix} -40 & 20 \\ 20 & -10 \end{bmatrix} v = 0$$

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$$R_2 \rightarrow 2R_2 + R_1$$

13.

$$\begin{bmatrix} -40 & 20 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\Rightarrow -40 v_1 + 20 v_2 = 0$$

$$\Rightarrow 40 v_1 = 20 v_2$$

$$\Rightarrow 2 v_1 = v_2$$

$$v_2 = 2$$

$$\Rightarrow v_1 = 1$$

First eigen vector:

Normalize:

$$\frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

For $d_2 = 0$.

$$(A^T A - dI) v = 0$$

$$\Rightarrow \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} v = 0$$

$$R_2 \rightarrow R_2 - 2R_1$$

14.

$$\Rightarrow \begin{bmatrix} 10 & 20 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\Rightarrow 10v_1 + 20v_2 = 0$$

$$\Rightarrow v_1 = -2v_2$$

eigenvector: $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

Normalized:

$$\frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \checkmark \quad \text{--- } (R_1)$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

~~15~~

Step 5:- Find the left singular vectors (columns of V).

$\xrightarrow{15}$
first colⁿ of V .

$$u_i = \frac{Av_i}{\sigma_i}$$

$$Av_1 = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 5 \\ 15 \end{bmatrix}$$

$$\boxed{Av_1 = \sqrt{5} \begin{bmatrix} 1 \\ 3 \end{bmatrix}}$$

$$u_1 = \frac{\sqrt{5} \begin{bmatrix} 1 \\ 3 \end{bmatrix}}{5\sqrt{2}} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$V \rightarrow$ orthogonal

$\hookrightarrow$ cols u_1 & u_2

since u_2 is orthogonal to u_1 .

Assume, $u_2 = \underbrace{(a, b)}^{15}$

$$a + 3b = 0$$

16.

$$u_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{-3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} \quad \checkmark$$

Step 6: - Write final SVD!

$$U = \begin{bmatrix} 1/\sqrt{10} & -3/\sqrt{10} \\ 3/\sqrt{10} & 1/\sqrt{10} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 5\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

$$A = U \Sigma V^T$$

17.

$$= \begin{bmatrix} \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} 5\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

Q.2 (Q.2):

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$$

Step 1: compute $A^T A$.

$$A^T A = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

Step 2:- Find eigenvalues of $A^T A$.

$$|A^T A - \lambda I| = \begin{vmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = (5-\lambda)^2 - 9 = 0$$

$$\Rightarrow \lambda^2 + 25 - 10\lambda - 9 = 0$$

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$$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$$

18.

$$\Rightarrow \lambda^2 - 8\lambda - 2\lambda + 16 = 0$$

$$\Rightarrow \lambda(\lambda - 8) - 2(\lambda - 8) = 0$$

$$\Rightarrow (\lambda - 2)(\lambda - 8) = 0$$

$$\Rightarrow \boxed{\lambda = 8, 2}$$

Step 3:- Singular values:

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{8} = 2\sqrt{2}$$

$$\sigma_2 = \sqrt{\lambda_2} = \sqrt{2}$$

So,

$$\Sigma = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}.$$

Step 4:- Eigenvectors of $A^T A$.

For $\lambda_1 = 8$.

$$(A^T A - 8I) \mathbf{x} = 0$$

$$\Rightarrow \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \mathbf{x} = 0$$

$$\Rightarrow \begin{bmatrix} -3 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

19.

$$\Rightarrow -3x_1 + 3x_2 = 0$$

$$\Rightarrow 3x_1 = 3x_2$$

$$\Rightarrow x_1 = x_2$$

First eigenvector: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Normalize: $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow \underline{\underline{\text{First or of } V.}}$

For $\lambda_2 = 2$:

$$(A^T A - 2I)x = 0$$

$$\Rightarrow \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 3 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

19

$$\Rightarrow 3x_1 = -3x_2$$

20.

$$\Rightarrow \boxed{x_1 = -x_2}$$

$$x_2 = -1$$

$$\boxed{x_1 = 1}$$

eigenvector: $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Normalize: $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow$ second colⁿ of V .

$$V = \begin{bmatrix} \textcircled{v_1} & \textcircled{v_2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

step 5:- Compute $U = \frac{AV}{\Sigma}$

~~First~~ $u_i = \frac{Av_i}{\sigma_i} \rightarrow$ ith colⁿ of V .

$$Av_1 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

21.

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 4/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}$$

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Next: $Av_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ -\sqrt{2} \end{bmatrix}$$

$$u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Step 6:

Final form of SVD.

$$U = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}, \quad V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$A = U \Sigma V^T$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

Ans

Positive Definite:

23. Every quadratic function of the form $ax^2 + 2bxy + cy^2$ stationary point at $(0,0)$.

Defⁿ:- A function that vanishes at point $(0,0)$ and is strictly positive at other points is called a positive definite function.

eg:- $f(x,y) = x^2 + y^2$.

* Symmetric matrices can be used to define quadratic form.

$$f(x) = \underline{x^T A x} \rightarrow \text{quadratic form.}$$

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 6 \end{bmatrix}$$

$$\underbrace{x^T A x}_{\hookrightarrow} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

2x2 2x1

24.

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 + 3x_2 \\ 3x_1 + 6x_2 \end{bmatrix}$$

$$= x_1^2 + 3x_1x_2 + 3x_1x_2 + 6x_2^2$$

$$= x_1^2 + 6x_1x_2 + 6x_2^2$$

→ quadratic form.

positive definiteness:-

(All eigenvalues are positive)

A symm. matrix A is positive definite if

$x^T A x > 0$ for all $x \neq 0$.

positive semi-definite:- $x^T A x \geq 0$ (Non-negative)

First order partial derivatives

25.

$$f_x = \frac{\partial f}{\partial x}, \quad f_y = \frac{\partial f}{\partial y}$$

→ 2nd order partial derivatives

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}$$

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$$

At a stationary point: → First order derivative vanishes.

$$f_x = 0, \quad f_y = 0.$$

$$\rightarrow D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - f_{xy}^2$$

Stationary points

Condⁿ

26.

Minima $\rightarrow$

$$\underline{f_{xx} > 0}, \underline{D > 0} \checkmark$$

Maxima $\rightarrow$

$$\underline{f_{xx} < 0}, \underline{D > 0} \checkmark$$

Saddle

$$D < 0$$

Inconclusive

$$D = 0$$

For a 2×2 real symmetric matrix

$$f(x, y) = v^T A v, \quad A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}, \quad v = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{pos. def.} \quad \left\{ \begin{array}{l} \underline{f(x, y) > 0} \\ \underline{a > 0, \quad \underline{ac - b^2 > 0}} \end{array} \right\}$$

$$\text{p.s.d.} \quad \left\{ \begin{array}{l} f(x, y) \geq 0 \\ a > 0, \quad ac - b^2 = 0 \end{array} \right\}$$

$$\text{neg. def.} \quad \left\{ \begin{array}{l} f(x, y) < 0 \\ a < 0, \quad ac - b^2 > 0 \end{array} \right\}$$

$$\text{n.s.d.} \quad \left\{ \begin{array}{l} f(x, y) \leq 0 \\ a < 0, \quad ac - b^2 = 0 \end{array} \right\}$$

$$\text{Indefinite} \quad \left\{ \begin{array}{l} f(x, y) > 0, \quad f(x, y) < 0 \\ ac - b^2 < 0 \end{array} \right\}$$

eg: -

27.

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

$$8 - 4 > 0$$

$$x = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x^T A x = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 4x + 2y \\ 2x + 2y \end{bmatrix}$$

$$= 4x^2 + 2xy + 2xy + 2y^2$$

$$f(x, y) = x^T A x = 4x^2 + 4xy + 2y^2$$

$$f_x = 8x + 4y, \quad f_y = 4x + 4y$$

$$8x + 4y = 0 \quad ; \quad 4x + 4y = 0$$

$$f_{xx} = 8, \quad f_{yy} = 4$$

$$f_{xy} = 4$$

~~x~~ (0, 0)
↑
stationary point

28.

$$D = f_{xx} \cdot f_{yy} - f_{xy}^2$$

$$D = 32 - 16 > 0$$

$$\delta a_n > 0$$

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

For a 2×2 matrix -

$$\lambda^2 - (\text{trace}) \lambda + (\det A) = 0$$

$$\text{Trace} = 4 + 2 = 6$$

$$\det A = 8 - 4 = 4$$

$$\Rightarrow \lambda^2 - 6\lambda + 4 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 16}}{2} = \frac{6 \pm \sqrt{20}}{2}$$

29.

$$d = 3 \pm \sqrt{5}$$

$$d_1 = 3 + \sqrt{5}, \quad d_2 = 3 - \sqrt{5}$$

$$\underline{d_1 > 0}$$

$$d_2 > 0$$

30.

31.

32.

33.

34.

35.